

Design of a 4-port Half-Circular AVA MIMO Antenna for Efficient Radar and Sensing Applications

Md. Arif Islam¹, Mohammad Habibur Rahman Razib², Md. Farhad Hossain³, Debprosad Das⁴, Md. Azad Hossain⁵

Department of Electronics and Telecommunication Engineering^{1,2,3,4,5}

Chittagong University of Engineering and Technology^{1,2,3,4,5}

¹u1908038@student.cuet.ac.bd, ²razib15ete@gmail.com, ³farhad.hossain, ⁴u19mete024p, ⁵azad}@cuet.ac.bd

Abstract—The paper described the concept and technological assessment of a 4-port half-circular antipodal Vivaldi antenna (AVA) MIMO configuration that was suitable for radar and sensing applications. The antenna was constructed using a Teflon substrate ($\epsilon_r = 2.1$, $\tan\delta = 0.0002$) and a 50- Ω microstrip line, which operated in the frequency range of 13 GHz to 40 GHz. Due to the implementation of the ground circle and flare gaps, isolation levels better than -28 dB were achieved, resulting in a gain of 10.5 dB, which was acceptable and met the design criteria. The envelope correlation coefficient (ECC) was 0.0002, indicating that there was minimal mutual coupling among the elements. The efficiency of the system was further elucidated through the high diversity gain (DG) of 9.998 dB, channel capacity loss (CCL) not exceeding 0.4 bits/s/Hz, total active reflection coefficient (TARC) being less than -10 dB, mean effective gain (MEG) varying between -3.6 dB and -3.01 dB, and MEG ratio ranging between -0.07 dB and 0.089 dB. The performance parameters of the antenna suggested that it was suitable for good radar and sensor instances. Some targeted applications included wide coverage and high gain due to accurate targeting and mapping of images of the surrounding environment.

Keywords— Antipodal Vivaldi Antenna (AVA), Multiple Input Multiple Output (MIMO), Wideband antenna, ECC, DG, CCL, TARC, MEG, MATLAB.

I. INTRODUCTION

Radar and sensor systems are essential technology in contemporary sectors including aerospace, automotive, military, and industrial automation. In order to provide high-resolution detection, real-time environmental monitoring, and dependable data collecting in a variety of settings, these systems rely on sophisticated antenna designs [1]. The development of innovative antenna systems is essential to achieve improved performance metrics as the market for more dependable, high-performance and efficient systems grows [2]. The performance of traditional antennas in contemporary applications is hampered by a number of restrictions. They frequently have small bandwidths, which restricts their capacity to process a large variety of frequencies. They could also have less directional radiation efficiency and inadequate gain, which would lessen overall efficacy and erode signal strength [3]. MIMO (multiple input multiple output) antennas have emerged as viable options for these applications due to their intrinsic wideband capabilities, high gain, decrease of mutual coupling, and directional radiation properties, making them appropriate for radar and

sensing operations [4]. The Vivaldi MIMO antenna is appropriate for this purpose among other types of MIMO antennas [5]. The Vivaldi antenna is perfect for radar systems that need focussed beams for accurate object identification because of its end-fire radiation pattern, which channels energy along the antenna's axis [6]. Its great angle coverage from its horizontal radiation improves its capacity to identify objects across a large region [7].

In recent work, a number of methods have been put out to enhance MIMO antenna performance measures, including isolation, mutual coupling, return loss, and directional radiation characteristics. In [8], an F-shaped stub is reported to reduce the mutual coupling of the MIMO antenna. But its frequency range is too small. In [9], a four-element Vivaldi MIMO antenna measuring 216×216 mm² is presented, featuring tunable band-notch functionality through varactor-loaded SRRs (Split-Ring Resonator) and adjustable narrowband options. Where the size is too large. For better MIMO performance, reducing mutual coupling and attaining port isolation of less than -20 dB is vital. This level of isolation leads to improved MIMO system metrics [10]. In [11], EBG structures are employed between array elements to notably enhance isolation. Conversely, [12] presents a compact dual-band-rejected MIMO Vivaldi antenna (26×28 mm²) featuring T-slots and SRRs, which achieves a relatively low isolation of -15 dB. In [13] proposes a compact 4-port orthogonal MIMO antenna with embedded slits but achieves inadequate isolation of -17.5dB. Lastly in [14], a 1×2 MIMO Palm Tree Co-planar Vivaldi antenna was simulated over a frequency range of 0.5–4.5 GHz. However, the design is notably large and costly.

In this paper, a half-circular 4-port Vivaldi antenna is designed to achieve 180-degree horizontal pattern diversity in the frequency range of 13 GHz to 40 GHz, which provides the perfect combination of high resolution, wide bandwidth, environmental adaptability, and compatibility with modern radar and sensing applications. Initially, a single-element antipodal Vivaldi antenna is implemented. A bending feed technique is then applied to both the main and side antennas to form a MIMO system. To enhance isolation a ground circle is introduced. Additionally, flare gaps between the elements are incorporated to further improve isolation. This design enhances key MIMO performance metrics, such as envelope

correlation coefficient (ECC), diversity gain (DG), channel capacity loss (CCL), total active reflection coefficient (TARC), and mean effective gain (MEG), surpassing the results of previous studies in the literature.

II. DESIGN AND ANALYSIS

The proposed Vivaldi MIMO antenna structure, along with its dimensions is simulated and analyzed using CST. Designing a 4-port MIMO antenna system with a half-circle shape involves integrating antipodal Vivaldi antennas, each covering a 45-degree sector. Given that the half-circle structure has four sides, the antennas need to be strategically arranged to ensure comprehensive 180-degree coverage. Because of the radar and sensing applications, where wide-angle coverage is essential. Each of the 4 ports delivers a 45-degree beam. The design process starts with a single antipodal Vivaldi antenna, which is configured with dimensions of $36 \times 50.51 \text{ mm}^2$ to ensure a 45-degree coverage. It employs copper annealed as both conductor and ground, with a thickness of 0.035 mm, and is mounted on a Teflon (polytetrafluoroethylene, PTFE) substrate that is 0.8 mm thick, featuring a relative dielectric constant (ϵ_r) of 2.1 and a loss tangent ($\tan\delta$) of 0.0002. The geometrical parameters of the single AVA element are shown in Fig. 1, while the optimized values of these parameters are provided in Table I. Additionally, the optimized exponential equation for the curved flare profile is provided below.

$$y_1 = e^{qx} + a; \text{ where } 0 \leq x \leq 27 \quad (1)$$

$$y_2 = e^{rx} - b; \text{ where } 0 \leq x \leq 10 \quad (2)$$

Here, y_1 , y_2 denoted as the curved profile of the flare, while q and r represent the tapering rates for controlling the curved profile. The constant optimized values a and b are utilized to determine the appropriate width of the feedline, ensuring impedance matching and efficient power transfer between components. To match the 50Ω microstrip line, the feedline width is adjusted to 2.5 mm.

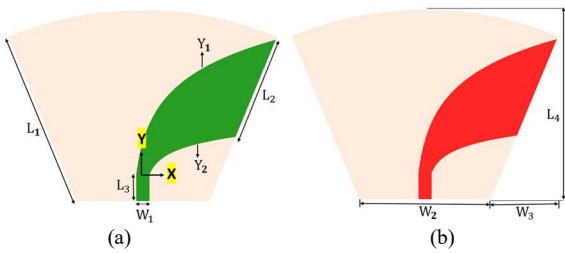


Fig. 1. Structure of the proposed single-element antipodal Vivaldi antenna (AVA) (a) front view where the flare (green) acts as a radiating plane (b) back view where the flare (red) acts as a ground plane.

Following the optimization of the single-element AVA, a 4-port AVA MIMO antenna has been developed to cover a 180-degree sector as depicted in Fig. 2. A bending feed technique is applied to the main and side antennas, incorporating a bending radius of (r_1) 13.26 mm and (r_2) 83 mm to ensure consistent feed width across the design.

In Fig. 3, a ground circle was used to enhance the isolation and return loss. After that, as shown in Fig. 4, the 4-port AVA MIMO design was improved for better isolation by introducing flare gaps between the two AVA components.

TABLE I. OPTIMIZED VALUES OF THE PROPOSED ANTENNA'S GEOMETRICAL PARAMETERS.

Parameter	Value (mm)	Parameter	Value (mm)	Parameter	Value (mm)
L_1	33.5	W_3	12.83	a	2.25
L_2	19.8	W_4	3	b	.25
L_3	5	r_1	13.26	q	0.13
L_4	36	r_2	83	r	0.4
W_1	2.5	r_3	32	L_5	3
W_2	24.85	r_4	66		

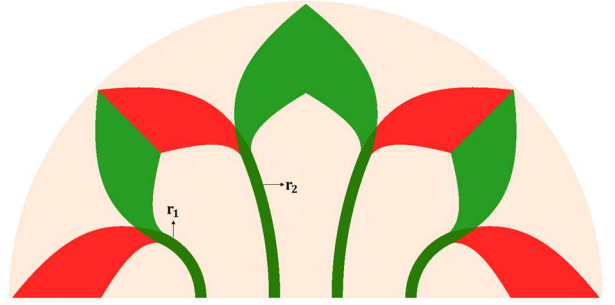


Fig. 2. Transparent structure of 4-port AVA MIMO-1.

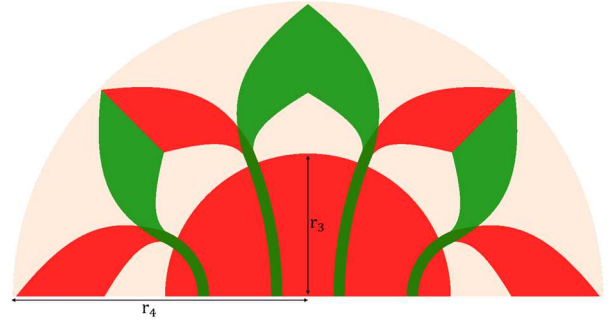


Fig. 3. Transparent structure of 4-port AVA MIMO-2 (with a half-circular ground plane).

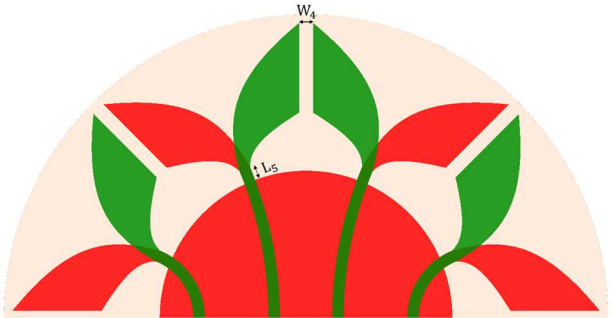


Fig. 4. Transparent structure of proposed 4-port AVA MIMO-3 (including half-circular ground plane and flare gaps between two AVAs).

III. SIMULATED RESULTS

This section provides a thorough examination of the simulated outcomes and evaluations of the AVA MIMO configurations, encompassing various design types in addition to the single element AVA, the suggested 4-port AVA MIMO without a ground circle and no flare gaps between two AVAs, with a ground circle and no flare gaps between two AVAs and finally with ground circle and flare gaps between two AVAs. Reflection coefficient, isolation, gain, efficiency, radiation pattern, surface current

distribution, envelope correlation coefficient (ECC), diversity gain (DG), channel capacity loss (CCL), total active reflection coefficient (TARC), and mean effective gain (MEG) are among the parameters that are compared.

A. Single element AVA Results

Over the frequency range of 13 GHz to 40 GHz, the single-element AVA antenna's estimated reflection coefficients regularly fall below -10 dB, reaching a minimum of -38 dB at 37.6 GHz seen in Fig. 5. As seen in Fig. 6, the antenna's simulated gain fluctuates between 7 dB and 10.5 dB, peaking at 10.5 dB at 38 GHz. Additionally, it maintains a high radiation efficiency of around 97% across the frequency range, attributed to the low-loss Teflon substrate with a loss tangent of 0.0002, resulting in excellent gain and efficiency performance.

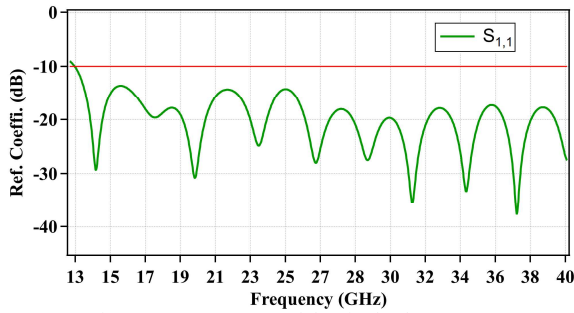


Fig. 5. $S_{1,1}$ parameter of the single element AVA.

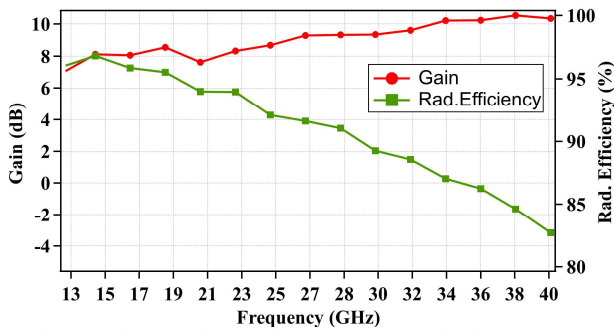


Fig. 6. Gain and radiation efficiency of the single element AVA.

B. MIMO AVA Results

In a 4-port MIMO antenna system, the middle two ports and the outer two ports of the antenna are symmetrically designed for balanced performance. The feedline is bent into the desired configuration to ensure proper impedance matching. In Fig. 7, the reflection coefficient and isolation across the frequency range of 13 GHz to 40 GHz are well-controlled, with S-parameters below -10 dB and isolation levels better than -23 dB.

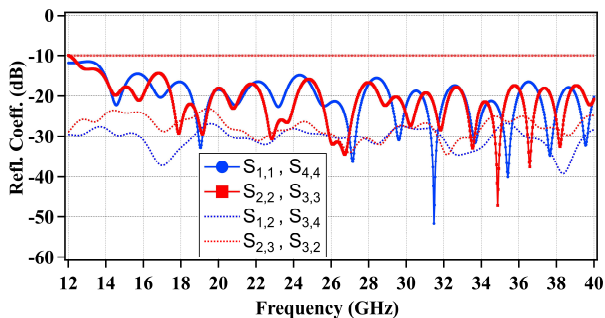


Fig. 7. S-parameters of the 4-port AVA MIMO-1.

To further improve isolation among the 4-port MIMO antennas, a half-circular ground plane circle was introduced into the lower flare surface with 0.035 thickness which can maintain lower levels of mutual coupling, improving isolation and overall system performance resulting in the isolation of below -25 dB, as shown in Fig. 8.

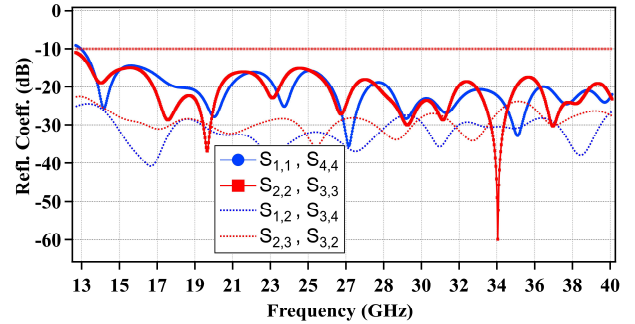


Fig. 8. S-parameters of the 4-port AVA MIMO-2.

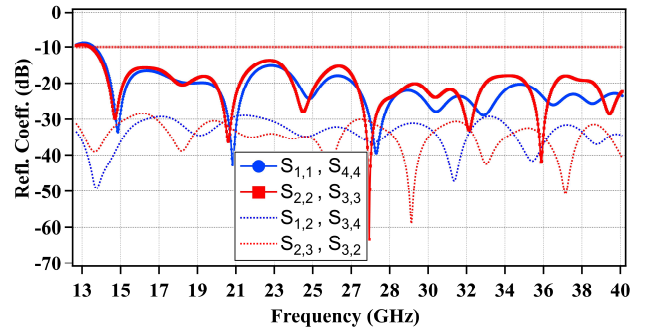


Fig. 9. S-parameters of the proposed 4-port AVA MIMO-3

To improve again the isolation among the 4-port MIMO antennas, flare gaps were introduced between two adjacent antipodal Vivaldi antennas (AVAs), resulting in the isolation of below -28 dB, as shown in Fig.9. The antenna shows a gain varying between 8 dB and 10.5 dB, with the peak gain occurring at 37.5 GHz in Fig.10.

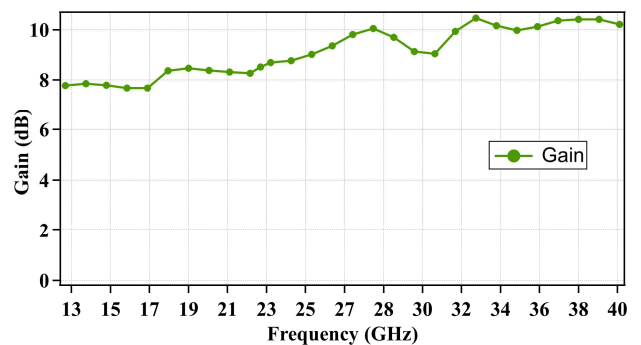


Fig. 10. Gain of the proposed 4-port AVA MIMO-3.

Important features of the surface current distribution become apparent in the half-circular 4-port AVA MIMO system operating at 31 GHz when ports 1 and 2 are energized, as shown in Fig.11. There is symmetry in the locations of ports 1 and 2 with respect to ports 4 and 3. Upon activation, the currents primarily flow along the feed line before igniting and expanding across the surface of flare structures. In order to ensure that the currents are primarily contained in each of the individual antennas, the 3 mm gap between the elements

also aids in reducing mutual coupling. This decrease in mutual coupling enhances isolation between the ports, reducing interference and signal degradation. As a result, this design modification improves overall antenna performance because each port can now operate more independently.

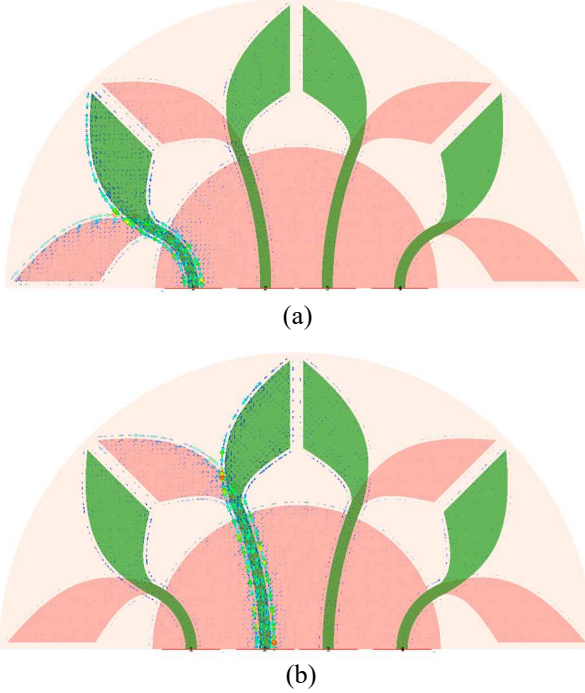


Fig. 11. Surface Current distribution of the proposed 4-port AVA MIMO-3 when ports 1 and 2 excited at 31 GHz frequency (a) Port 1 (b) Port 2.

Fig. 12 and 13 display the half-circular 4-port Vivaldi MIMO antenna's radiation pattern at 14 GHz and 31 GHz. The pattern shows how the antenna may deliver excellent range coverage and homogenous signal dispersion in all horizontal directions. Its large beamwidth and high gain result in a strong and consistent signal strength. One of the design's stronger points is the pattern diversity, which consists of four distinct radiation patterns for each of the four ports, each spaced 30 degrees apart. This configuration significantly reduces mutual coupling and interference between the ports, which leads to clearer signal transmission and greater overall efficiency. These characteristics make the antenna ideal for high-frequency applications requiring consistent, wide signal coverage, which improves the effectiveness of radar and sensing systems.

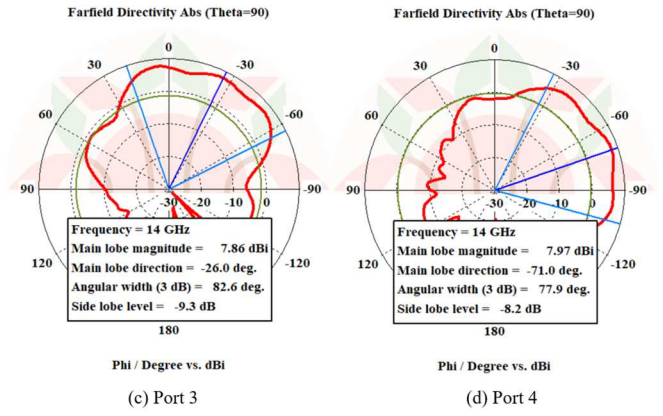


Fig. 12. Radiation pattern of the proposed half-circular shaped 4-port AVA MIMO-3 at 14 GHz frequency.

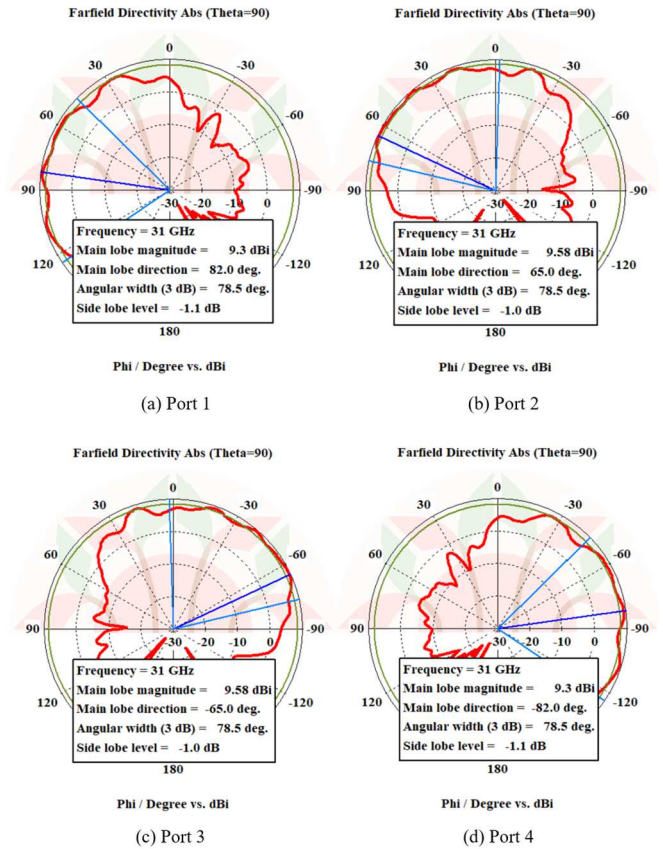
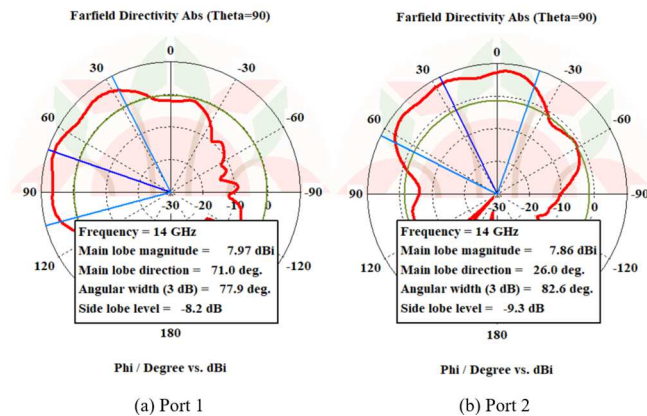


Fig. 13. Radiation pattern of the proposed half-circular shaped 4-port AVA MIMO-3 at 31 GHz frequency.

C. MIMO AVA performance characteristics

Many important factors must be considered to ensure that MIMO (Multiple Input Multiple Output) antenna designs work well. These include the channel capacity loss (CCL), which shows how much data capacity is lost, the envelope correlation coefficient (ECC), which shows how much the signals from different antennas interfere with one another, and the diversity gain (DG), which measures how much the system boosts signal strength under various circumstances and the antennas' ability to receive signals and compare their performance under various conditions is evaluated by the MEG, MEG Ratio and the TARC, which shows how much power is reflected the system.



The ECC is an important indicator for MIMO antenna performance, which shows the correlation between signals from different antenna elements. ECC values below 0.5 suggest good spatial diversity and minimal signal interference. As shown in Fig. 14, the maximum ECC of 0.0002 is far below this threshold, reflecting excellent signal independence and improved system performance. The ECC values for both ECC₁₂ and ECC₃₄ are identical, so only ECC₁₂ is plotted.

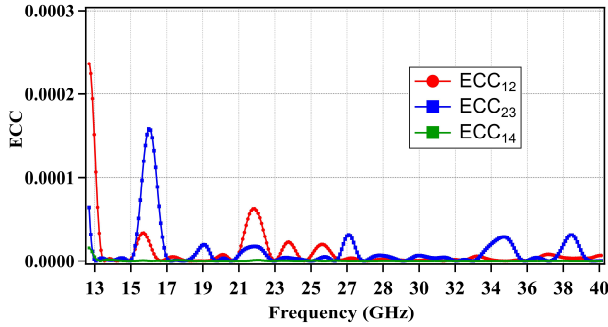


Fig. 14. Envelope correlation coefficient of the proposed 4-port AVA MIMO-3.

Diversity gain (DG) reflects how well a MIMO antenna system improves signal quality by minimizing fading and interference. A DG value of 10dB is optimal. In Fig. 15, the DG is 9.998 dB, indicating that the antenna performs very close to the ideal, effectively enhancing signal quality and reliability. The DG values for both DG₁₂ and DG₃₄ are identical, so only DG₁₂ is plotted.

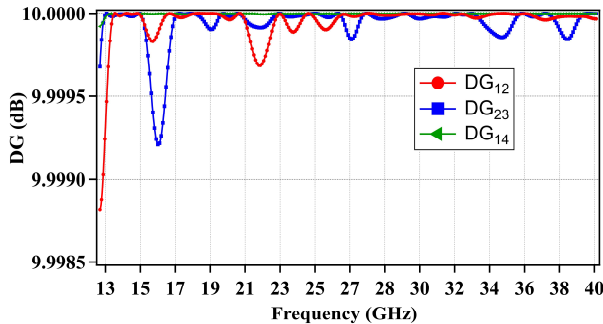


Fig. 15. Diversity gain of the proposed 4-port AVA MIMO-3.

Channel capacity loss (CCL) measures how much mutual coupling and interference in a MIMO system impact data rates, with 0.4 bits/s/Hz being the standard acceptable value. Fig. 16 shows that when the CCL is below 0.4, the system effectively minimizes capacity loss while maintaining strong performance.

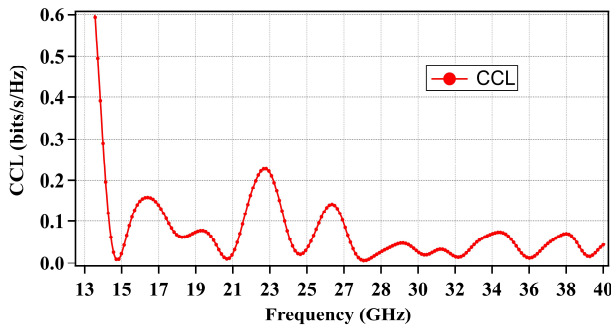


Fig. 16. Channel capacity loss of the proposed 4-port AVA MIMO-3.

The total active reflection coefficient (TARC) indicates how much power is reflected compared to the incoming power in a MIMO system, assessing how well it handles reflections. Fig. 17 shows TARC values below -10 dB across the frequency range, when phase difference is 0°, then the lowest TARC values is -24.9 dB, when 90° and 180°, the values are respectively -35.4dB and -55.2dB. Higher phase difference improves TRAC values and enhances the overall performance of MIMO parameters. This demonstrates that the system effectively reduces reflections and ensures strong signal quality between the antenna elements.

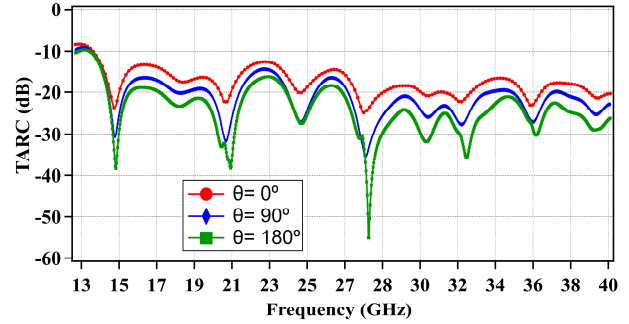


Fig. 17. Total active reflection coefficient of the proposed 4-port AVA MIMO-3 at different phase differences.

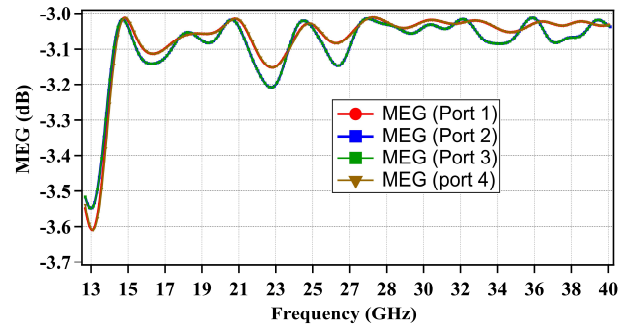


Fig. 18. Mean effective gain of the proposed 4-port AVA MIMO-3.

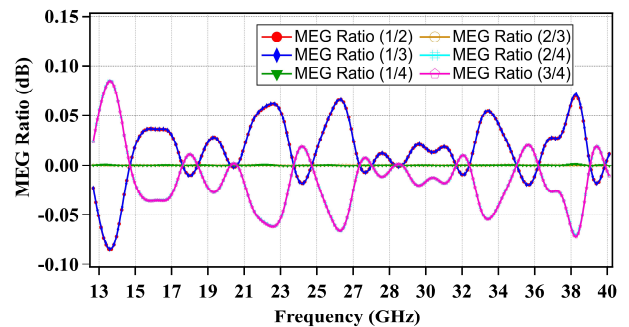


Fig. 19. MEG ratio of the proposed 4-port AVA MIMO-3.

Mean effective gain (MEG) represents the average gain of an antenna across different directions and conditions, with acceptable values typically between -3 dB to -12 dB Fig. 18 shows MEG values ranging from -3.6 dB to -3.01 dB, which falls within this range. Higher values closer to -3 dB indicate better performance, as the antenna maintains strong gain across various conditions.

The MEG ratio compares the mean effective gain between different antennas or configurations. An acceptable MEG ratio usually ranges from -3 dB to 3 dB. Fig. 19 shows that the MEG ratio ranges from -0.07 dB to 0.089 dB, well within the acceptable limits.

Table II shows the comparisons between the proposed antenna and recently published works, considering factors

TABLE II. PERFORMANCES COMPARISON WITH OTHER LITERATURE WORKS

Ref. No.	Size (mm ²)	No of Ant.	ϵ_r	BW (GHz)	Isolation (dB)	Gain (dB)	ECC	CCL (bits/s/Hz)	DG (dB)	TARC (dB)	MEG (dB)	MEG Ratio (dB)
[7]	50×30	2	4.4	2.5-14.5	<-20	4.1	<0.04	NM	>7.4	NM	NM	NM
[8]	216×216	4	2.2	3-10	<-20	8	<0.018	< 0.4	9.84	NM	NM	NM
[9]	85×85	2	4.4	3-12	<-20	5	0.009	0.14	9.97	NM	NM	NM
[10]	34×55.8	2	3.8	26-40	-30	12.8	NM	NM	NM	-20	NM	NM
[11]	26×28	2	2.6	2.7-10.9	-15	6.2	0.02	NM	NM	NM	NM	NM
[12]	40×40	4	2.2	3.3-3.6	-17.5	10	0.01	< 0.4	9.98	< -10	-15 to -6	0.97 to 1.03
[13]	36×36	4	2.2	24-32	-15	10.2	0.2	NM	NM	< -10	NM	NM
This work	132×66	4	2.1	13-40	<-28	10.5	0.0002	< 0.4	9.998	-55.2	-3.6 to -3.01	-0.07 to 0.089

*NM = Not Mentioned

IV. CONCLUSION

In this paper, three types of structures were proposed for performance enhancement through process optimization within the frequency range of 13 GHz to 40 GHz. Firstly, without flare gaps between two AVAs and without the ground circle, secondly without flare gaps between two AVAs and with a ground circle and thirdly with flare gaps and ground circle. These designs, featuring Teflon substrates and symmetrical 4-port configurations, among these three designs AVA MIMO-3 ensured efficient energy distribution, high isolation below -28 dB, and low mutual coupling. It achieved a gain of 10.5 dB and showed strong MIMO performance with a low ECC of 0.0002, a high DG of 9.998 dB, a CCL under 0.4 bits/s/Hz, and a TARC below -10 dB. Both the MEG and MEG ratios were within acceptable limits, confirming the antenna's reliability and suitability for efficient Radar and Sensing applications.

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